Youngmin Jeong

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Research Interests

Computer Vision, Robot Vision, Visual Perception, Multi-modal Learning, Vision-Language Models

Education

Seoul National University of Science and Technology (SeoulTech) Seoul, Korea
B.S. in Applied Artificial Intelligence, GPA: 3.9/4.5, Expected Graduation: Feb. 2027 *Mar. 2021 – Present*

Research Experience

Intelligent Pattern Analysis and Learning (IPAL) Lab Seoul, Korea
Undergraduate Researcher (Advisor: Prof. Beom-Seok Oh) *Oct. 2024 – Aug. 2026*

Artificial Intelligence for Remote Sensing (AIR) Lab Seoul, Korea
Undergraduate Researcher (Advisor: Prof. Seonyoung Park) *Apr. 2021 – Feb. 2023*

Publications

⟨ **International Journal** ⟩ (*Equal contribution is denoted by "*".*)

[1] Yerang Lee*, **Youngmin Jeong***, Beom-Seok Oh, Seonyoung Park, and Eun Gyung Chung, "A Deep Semantic Segmentation System for Detecting Multi-Scale Barns of Diverse Livestock Breeds in Satellite Imagery," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS)*, 2026. (*Under Review*)

⟨ **Domestic Conferences** ⟩

[1] **Youngmin Jeong** and Beom-Seok Oh, "GLCFormer: Improving Livestock Facility Detection in Satellite Imagery Using Graph-Based Land-Cover Information Fusion," *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, 2026.

[2] Yerang Lee*, **Youngmin Jeong***, and Beom-Seok Oh, "HeteroBarnMapper: A Detection-to-Mapping System for Heterogeneous Livestock Facilities with Reliable Evaluation," *Summer Annual Conference of IEIE*, 2025. (*Oral Presentation*)

[3] **Youngmin Jeong***, Yerang Lee*, and Beom-Seok Oh, "DualLCFormer: Livestock Facility Detection using a Dual Encoder Architecture with Land Cover Fusion," *Summer Annual Conference of IEIE*, 2025.

[4] **Youngmin Jeong**, Kyungil Lee, and Seonyoung Park, "Landsat 9 based Local Climate Zone Mapping using CNN in Rome," *Fall Annual Conference of the Korean Association of Geographic Information Studies (KAGIS)*, 2022. (*Extended Abstract*)

Research Projects

Dataset Construction and Management for Training Satellite Imagery Foundation Models *Mar. 2026 – Present*
Korean Society for Aeronautical and Space Sciences (KSAS)

- **Role:** Foundation model implementation and evaluation using curated multi-source satellite imagery datasets.

Development of Monitoring Methods for Livestock Facility Detection *Nov. 2024 – Nov. 2025*
National Institute of Environmental Research (NIER)

- **Role:** Led model development and extensive experimentation for an end-to-end livestock facility detection pipeline, including post-processing, object-level evaluation, and field validation.

Selected Projects

RS-AutoLabel: Training-Free Open-Vocabulary Remote Sensing Auto-Labeling *Sep. 2025 – Jun. 2026*
Capstone Design Project, SeoulTech

- Developed an open-vocabulary remote sensing auto-labeling system with region consistency refinement and user-assisted exemplar prompting.

Open-Vocabulary VLM-Based Interactive Object Labeling Automation System *Sep. 2025 – Nov. 2025*
2025 Space Hackathon, Korea Aerospace Research Institute (KARI)

- Developed an interactive object labeling automation system using open-vocabulary vision-language models.

Awards & Honors

President's Award, 8th National Undergraduate Capstone Design Competition, The Society for Aerospace System Engineering (SASE) *Aug. 2026*

Special Award, 2026 Capstone Design Competition, SeoulTech *Jun. 2026*

Top Excellence Award (KARI President's Award), 2025 Space Hackathon *Nov. 2025*

AIS Scholarship (Full Tuition), SeoulTech *Mar. 2021 – Present*